



Carisurg

Assignment 16

Final Documentation

Due Date: July 14th, 2026.

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Setup & Data Decisions

Dataset

The analysis was conducted using **55,121 Emergency Department (ED) encounters** from **triage_cleaned_v1.csv**, which represents the canonical output of the Week 5 data preparation pipeline. This dataset served as the foundation for all baseline modelling and evaluation.

Random Seed

A random seed of **42** was used to ensure that all data splitting and model training procedures were reproducible.

Train/Test Split

The dataset was divided into **80% training** and **20% testing** using **stratified sampling** based on the Emergency Severity Index (ESI). Stratification preserved the original class distribution across both subsets, ensuring that each ESI level was proportionally represented.

- **Training set:** 44,096 encounters
- **Test set:** 11,025 encounters

Leakage Prevention

To prevent data leakage, **Disposition** and **Previous Disposition** were excluded from the feature set. Both variables are only known after a patient's Emergency Department encounter has concluded and would therefore provide information unavailable at the time of triage. Including either variable would artificially inflate model performance while making the results clinically unrealistic.

Arrival Mode Encoding

Arrival mode was transformed into a binary **ambulance arrival** indicator. Within this dataset, ambulance arrivals are represented by the lowercase value **ambulance**. A total of **18,561 encounters (33.7%)** involved patients arriving by ambulance.

Class Imbalance

The ESI classes are highly imbalanced, with **ESI-1 representing only 77 patients (0.14% of all encounters)**. To reduce bias toward the majority classes, both baseline classifiers were trained using **class_weight='balanced'**.

Without class balancing, the models would be heavily incentivised to prioritise overall accuracy by rarely, if ever, predicting ESI-1. Such behaviour would produce misleadingly high performance while failing to identify the most clinically critical patients.

Selected Features

Model development used the **top ten predictor variables** identified during the Week 5 feature selection process:

- Age
- Oxygen saturation (SpO₂) (**triage_vital_o2**)
- Chest pain (**cc_chestpain**)
- Shortness of breath (**cc_shortnessofbreath**)
- Suicidal presentation (**cc_suicidal**)
- Alcohol intoxication (**cc_alcoholintoxication**)

- Altered mental status (`cc_alteredmentalstatus`)
- Ambulance arrival (`arrival_ambulance`)
- Respiratory rate (`triage_vital_rr`)
- Heart rate (`triage_vital_hr`)

Primary Metric Justification

The primary evaluation metric for this study is ESI-1 recall, with macro-averaged recall used as the overall summary measure across all five Emergency Severity Index (ESI) classes.

The choice of ESI-1 recall is driven by clinical risk rather than statistical performance. ESI-1 patients require immediate life-saving intervention and include presentations such as cardiac arrest, major trauma, anaphylaxis, and profound shock. Failing to identify one of these patients (a false negative) has far greater consequences than a delayed or incorrect prediction for a lower-acuity patient. In cardiac arrest, for example, the probability of survival decreases by approximately 10% for every minute that resuscitation is delayed. A model that incorrectly classifies an ESI-1 patient as ESI-2 or ESI-3 risks directing that patient to a standard waiting queue instead of the resuscitation bay, making false negatives the most clinically significant error.

This clinical priority is reinforced by the distribution of the dataset. ESI-1 represents only 77 of the 55,121 encounters (0.14%), with just 16 ESI-1 patients present in the test set. Because of this extreme class imbalance, overall accuracy is a misleading measure of model performance. A classifier that never predicts ESI-1 would still achieve approximately 99.86% accuracy, despite failing to identify every critically ill patient.

For this reason, **macro-averaged** recall is preferred over weighted metrics as the primary summary measure. Weighted metrics are dominated by the majority classes—particularly ESI-3, which accounts for nearly half of all encounters—and can therefore mask poor performance on clinically important minority classes. In contrast, macro recall assigns equal importance to each ESI level, ensuring that performance on ESI-1 contributes as much to the evaluation as performance on the more common classes.

Although the **weighted F1-score** is reported to provide a complete picture of model performance, it is not the primary optimisation target. Instead, model evaluation prioritises the ability to identify the highest-acuity patients while maintaining balanced performance across all ESI levels.

Accordingly, the principal failure mode considered throughout this evaluation is a false negative on ESI-1, where a patient requiring immediate resuscitation is incorrectly classified as a lower-acuity case. From a clinical perspective, this represents the most serious error a triage decision-support system can make.

Results

Head-to-Head Comparison

Table 1 compares the performance of the Dummy Classifier, Logistic Regression, and Decision Tree across the selected evaluation metrics.

Logistic Regression emerged as the strongest baseline model across the clinically meaningful metrics. It achieved the highest **macro recall (0.417)**, **ESI-1 recall (0.750)**, **weighted F1-score (0.327)**, and **macro F1-score (0.214)**. Although its overall accuracy (25.4%) appears modest, this result is expected because the model was optimised to improve performance on minority classes rather than maximise overall accuracy.

Most importantly, Logistic Regression correctly identified **12 of the 16 ESI-1 patients** in the test set. By comparison, the Dummy Classifier achieved an ESI-1 recall of approximately **0.14**, correctly identifying only two patients through random chance. This substantial improvement indicates that the selected clinical features contain meaningful predictive information rather than random statistical variation.

The Decision Tree produced similar overall accuracy (25.5%) but consistently underperformed Logistic Regression on the metrics most relevant to clinical decision support. Its lower macro recall (0.386) and ESI-1 recall (0.563) suggest that the tree struggled to learn robust decision boundaries for the minority classes under the imposed depth constraint.

Taken together, these findings suggest that a simple linear model provides a stronger baseline than a shallow Decision Tree for this dataset. Although neither model is suitable for clinical deployment,

Logistic Regression establishes a meaningful benchmark against which more sophisticated models can be evaluated in future work.

Model	Accuracy	Macro Recall	ESI-1 Recall	Weighted F1	Macro F1
Dummy (stratified)	0.375	0.204	~ 0.14	0.375	0.204
Logistic Regression	0.254	0.417	0.750	0.327	0.214
Decision Tree (depth=5)	0.255	0.386	0.563	0.302	0.170

Logistic Regression — Per-Class Results

The per-class evaluation reveals that Logistic Regression successfully prioritised the most clinically critical patients while struggling with overall class discrimination.

The strongest result was the **ESI-1 recall of 0.75**, indicating that the model correctly identified **12 of the 16 highest-acuity patients**. Although the corresponding precision was effectively zero because many non-ESI-1 patients were also predicted as ESI-1, this behaviour reflects the intended effect of class balancing. The model favoured sensitivity over precision, accepting more false alarms to reduce the risk of missing critically ill patients.

Performance declined substantially for the intermediate acuity levels. ESI-2 achieved a recall of **0.31**, while ESI-3 achieved **0.20**, indicating that many patients in these categories were confused with neighbouring ESI levels. This is not surprising, as adjacent triage categories often share

overlapping clinical characteristics, making them inherently more difficult to separate using only ten predictor variables.

Lower-acuity patients demonstrated mixed performance. ESI-4 achieved similar precision and recall (0.25 and 0.26, respectively), while ESI-5 recorded a relatively high recall (0.57) but extremely low precision (0.05). This suggests that the model frequently over-predicted the least urgent category, resulting in many false positives despite correctly identifying more true ESI-5 patients.

Overall, these results illustrate the central trade-off introduced by class balancing. The model substantially improved detection of rare, high-risk patients at the expense of precision and performance on the more common acuity levels. For an early-stage triage decision-support tool, this trade-off is clinically defensible, as failing to identify a critically ill patient is generally considered a more serious error than generating additional clinical reviews.

Class	Precision	Recall	F1-Score	Support
ESI 1	0.00	0.75	0.01	16
ESI 2	0.69	0.31	0.42	3,585
ESI 3	0.58	0.20	0.30	5,402
ESI 4	0.25	0.26	0.26	1,779
ESI 5	0.05	0.57	0.09	243

Decision Tree (max_depth=5) — Per-Class Results

The Decision Tree demonstrated a different pattern of behaviour from Logistic Regression. While it achieved slightly higher recall for the majority classes, its performance deteriorated considerably for the clinically important minority classes.

The most notable result was its **ESI-1 recall of 0.56**, meaning that only **9 of the 16 ESI-1 patients** were correctly identified. Compared with Logistic Regression, the tree missed three additional critically ill patients, reducing its suitability as a baseline for emergency triage support.

The most significant limitation, however, was its complete failure to predict **ESI-4**. Every one of the **1,779 ESI-4 patients** in the test set was assigned to another class, resulting in zero precision, zero recall, and a zero F1-score. This behaviour is consistent with shallow decision trees trained on highly imbalanced datasets. Under a maximum depth of five, the tree lacked sufficient complexity to learn decision boundaries that isolated the ESI-4 population while simultaneously accounting for class-balanced weighting.

Performance on ESI-2 and ESI-3 remained modest, with recalls of **0.40** and **0.22**, respectively. Although these values exceeded those achieved by Logistic Regression for ESI-2, they were accompanied by poorer performance on the highest-acuity patients, making the trade-off less favourable from a clinical perspective.

Overall, the Decision Tree appears to have underfit the dataset. Its restricted depth simplified the learned decision rules, limiting its ability to distinguish between clinically similar patient groups. Future work should investigate deeper trees, ensemble methods such as Random Forests or

XGBoost, or alternative weighting strategies to determine whether the observed limitations arise from the model architecture or the imposed complexity constraint.

Class	Precision	Recall	F1-Score	Support
ESI 1	0.01	0.56	0.01	16
ESI 2	0.50	0.40	0.44	3,585
ESI 3	0.57	0.22	0.32	5,402
ESI 4	0.00	0.00	0.00	1,779
ESI 5	0.04	0.75	0.08	243

LR Coefficient Sanity Check

For the ESI 1 class, a positive coefficient means the feature pushes the prediction toward ESI 1 (more urgent). Clinical expectations: chest pain, altered mental status, suicidal presentation, and ambulance arrival should all be positive; SpO₂ should be negative (lower oxygen = more urgent).

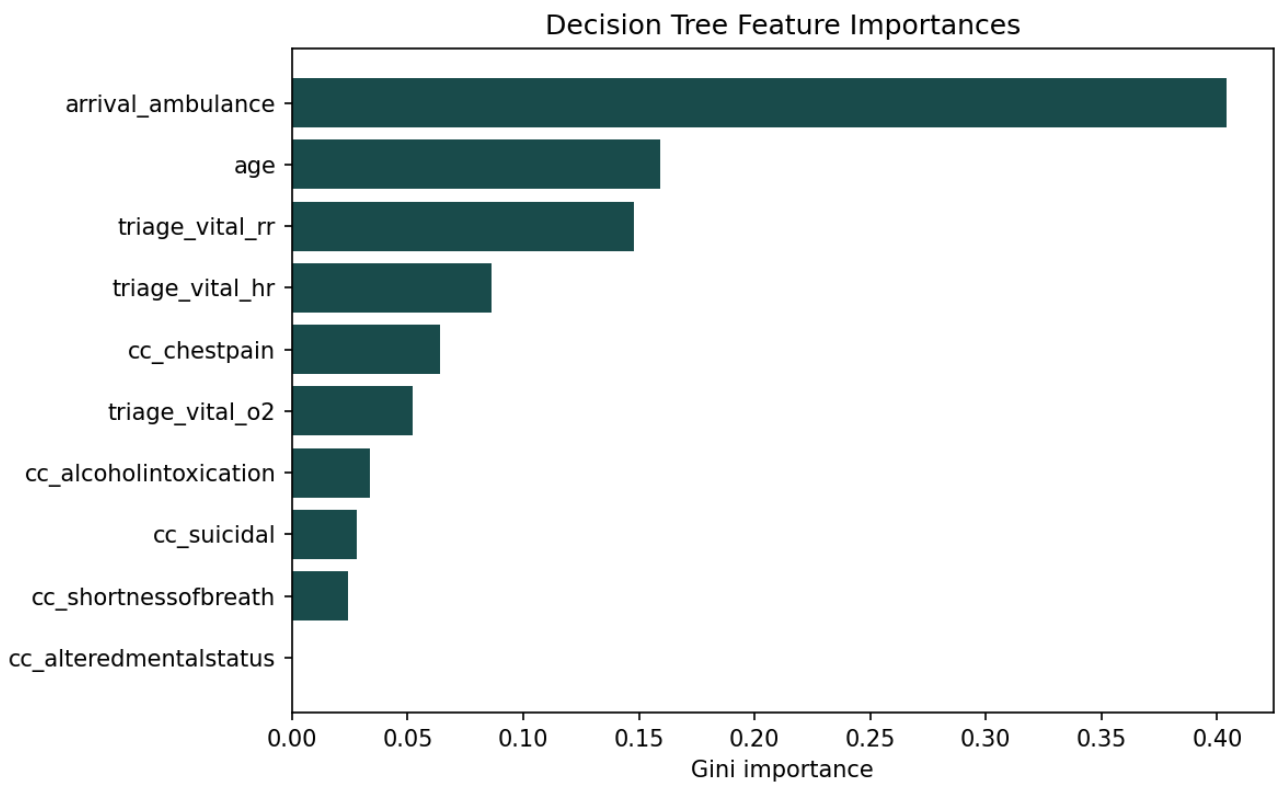


Figure 1. *Decision Tree Feature Importances*

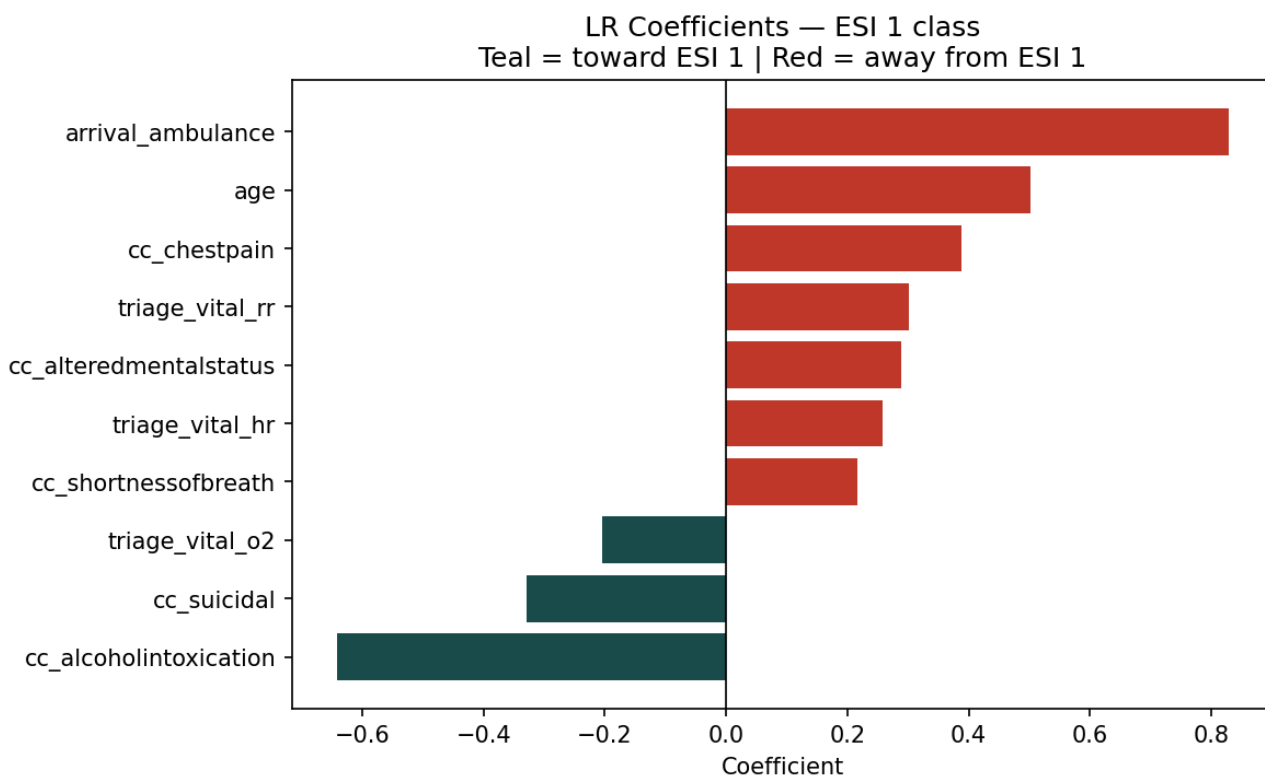


Figure 2. Logistic Regression Coefficients for the ESI 1 class (Teal = toward ESI 1, red = away from ESI 2)

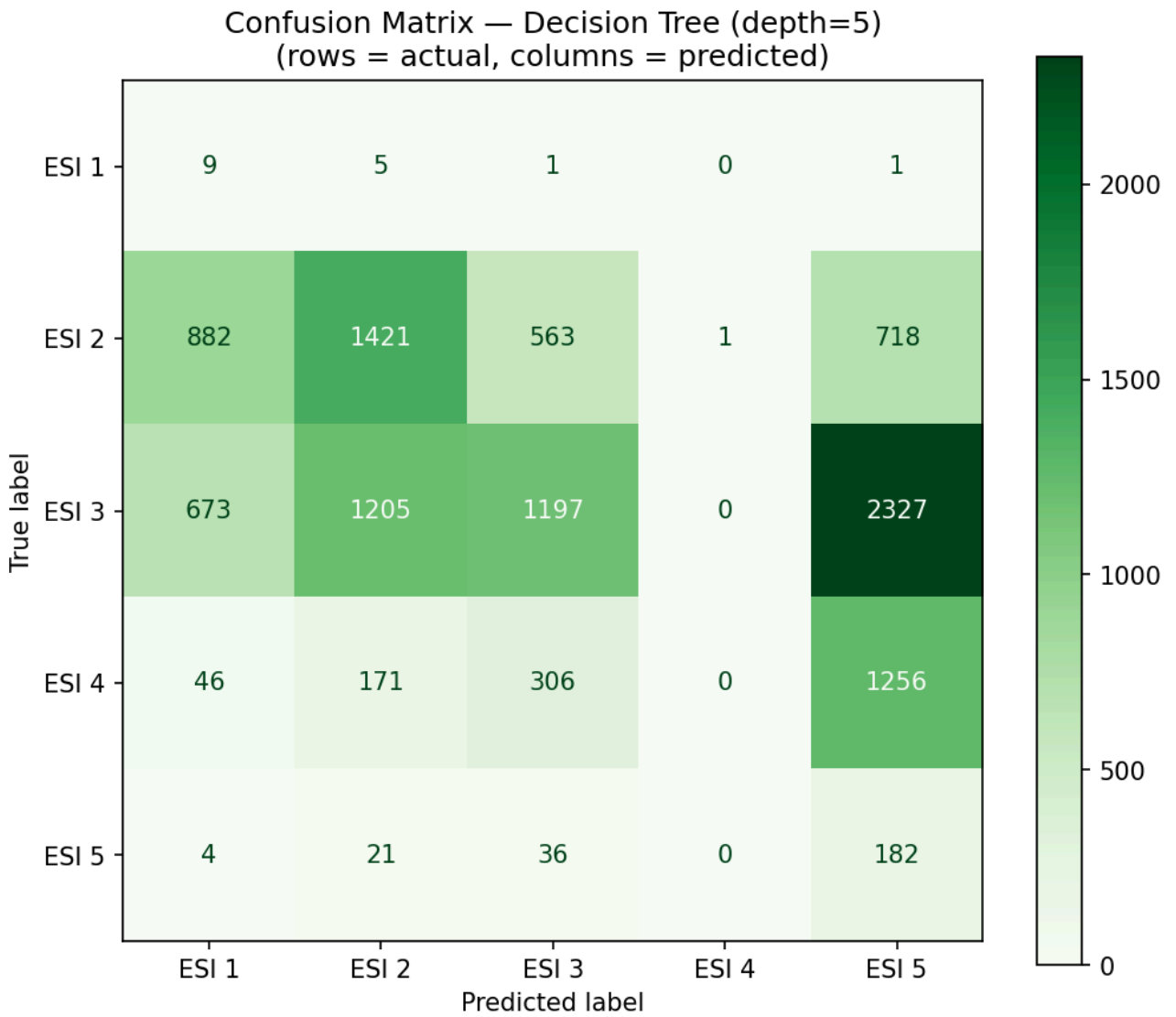


Figure 3. Confusion Matrix—Decision Tree (Depth = 5), (Rows = Actual, Columns = Predicted.)

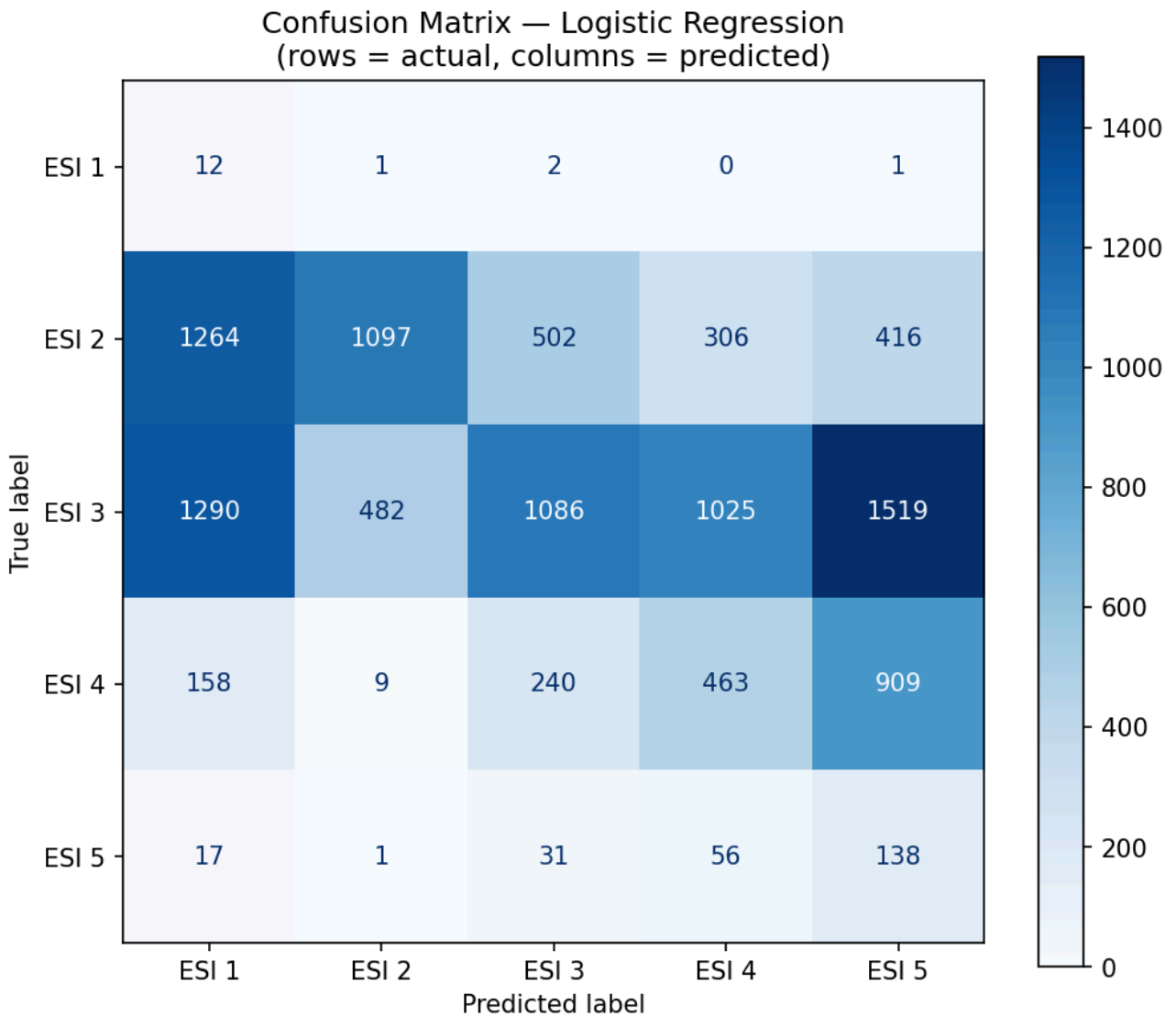


Figure 4, *Confusion Matrix — Logistic Regression. (Rows = Actual, Columns = Predicted)*

Failure Mode Reflection

Following the Tutorial 4 reflection framework — which group is affected, why the failure occurs, what the implications are, and the next step required — the following concerns were identified:

Primary Concern: ESI-1 False Negatives

The most critical failure identified is the Logistic Regression (LR) model missing 4 of the 16 ESI-1 patients, resulting in a 25% false negative rate. In a deployed triage environment, these patients would not be directed immediately to the resuscitation bay and would instead enter a waiting queue. For critically unstable patients, such as those experiencing cardiac arrest, this delay could result in catastrophic consequences.

This highlights that overall model performance cannot be evaluated solely through accuracy or average metrics. In emergency triage, the consequences of missing a high-acuity patient are significantly greater than those of misclassifying a lower-acuity patient. Improving sensitivity for ESI-1 cases will require further calibration, additional high-acuity data, and evaluation using safety-focused performance measures before deployment could be considered.

ESI-2 Recall Failure

The LR model also demonstrates limited performance for ESI-2 patients, achieving a recall of only 0.31. This group includes patients experiencing conditions such as stroke, sepsis, acute coronary syndrome (ACS), and severe pain who require timely physician assessment. Misclassifying these patients into other categories may result in delayed care or inefficient resource allocation.

This behaviour is partly a consequence of using `class_weight='balanced'`, which increases the influence of minority classes but can reduce precision among more common classes. While this approach attempts to address class imbalance, it introduces a trade-off between identifying rare critical cases and maintaining reliable predictions across all acuity levels. Before any deployment consideration, threshold calibration and additional optimisation are required to achieve a clinically acceptable balance.

Decision Tree ESI-4 Failure

The Decision Tree (DT) model completely fails to identify ESI-4 patients, assigning this category to zero patients. As a result, all 1,779 ESI-4 patients in the test dataset — representing semi-urgent cases requiring one resource — are misclassified. This suggests that the current depth-5 tree structure, combined with balanced weighting, is unable to adequately represent this class.

Although improving detection of rare high-acuity cases is important, a model that cannot recognise a substantial portion of the emergency department population would not be operationally useful.

Future experiments should investigate alternative tree depths, weighting strategies, feature engineering approaches, or different model architectures to improve the representation of this class.

Lack of Re-triage Capability

A further limitation is that the current classifier only performs a single prediction at the point of triage and does not account for patient deterioration over time. For example, a patient initially

classified as ESI-3 who deteriorates to ESI-1 within 30 minutes would not be automatically identified by the system.

This reflects the broader limitation of treating emergency triage as a static classification problem. In practice, patient conditions evolve, and clinical teams continuously reassess risk. Therefore, any AI-assisted triage tool must complement the nurse's clinical judgement and visual monitoring rather than replace it. Future development should consider mechanisms for dynamic reassessment and re-triage support.

ESI-1 Sample Size Caveat

The interpretation of ESI-1 performance is also limited by the small sample size available in the test set. With only 16 ESI-1 patients, each prediction changes recall by 6.25 percentage points. Therefore, while the observed recall value of 0.750 is directionally encouraging, it remains statistically fragile and should not be over-interpreted.

Additional high-acuity examples are necessary to validate whether the model is genuinely learning meaningful patterns or whether performance is influenced by the limited number of available cases.

Caribbean Dataset Mismatch

The largest limitation affecting all findings is the mismatch between this dataset and the intended deployment context. The dataset represents a single US hospital population and does not evaluate whether these models would transfer effectively to Caribbean emergency departments such as Mercer General.

Differences in patient populations, documentation practices, clinical workflows, available resources, and triage procedures may significantly influence model performance. Therefore, these results demonstrate model behaviour within the current dataset but cannot support deployment-oriented claims in a Caribbean setting.

Phase 1 — developing Caribbean-sourced emergency department data — remains the necessary prerequisite before evaluating AI-assisted triage in this context. Building locally relevant datasets will ensure that future models are trained and validated against the realities of the healthcare environments they are intended to support.

Macro vs Weighted F1

Weighted F1 calculates performance by weighting each class according to the number of patients it contains. Since ESI-3 patients represent 49% of the test set, the weighted F1 score is strongly influenced by performance on this majority class. As a result, a model that performs well on ESI-3 patients while failing to identify minority classes could still achieve a relatively high weighted F1 score despite being clinically inappropriate for triage support.

Macro F1 addresses this limitation by assigning equal importance to every class, regardless of its frequency. This means that ESI-1 patients, who represent only 0.14% of the dataset, contribute equally to the metric as ESI-3 patients, who represent 49%. Therefore, poor performance on critical minority classes, such as failing to identify ESI-1 patients, will be reflected in the macro F1 score rather than being masked by majority-class performance.

For an AI-assisted clinical triage system, ESI-1 recall is the key metric to monitor because missing high-acuity patients represents the greatest safety risk. Macro F1 provides a balanced summary of performance across all acuity levels, while weighted F1 is included as a supplementary measure to describe overall dataset-level performance. However, weighted F1 should not be treated as the primary optimisation target for deployment-oriented decision support.

Appendices

Supplementary Links

1. The Loom link can be found [here](#).
2. The GitHub link can be found [here](#)
3. Evidence for every figure above is included alongside this memo in the GitHub:
 - a. **figs/01-06_*.png**
 - b. **notebooks/Week6_Baseline_Model**

Glossary

A reference index for clinical, technical, and research terminology used throughout this proposal. Intended for any reviewer — clinical or non-clinical — who needs a quick definition without breaking flow.

Clinical & Vital Sign Terms

Term	Meaning
ED	Emergency Department
ESI	Emergency Severity Index — the 5-level triage scale used at Mercer General (1 = immediate, 5 = non-urgent)
SBP	Systolic Blood Pressure — the peak pressure during a heartbeat (top number in a BP reading)
DBP	Diastolic Blood Pressure — the pressure between heartbeats (bottom number in a BP reading)
MAP	Mean Arterial Pressure — average pressure driving blood to organs across a cardiac cycle; calculated as $(SBP + 2 \times DBP) / 3$
GCS	Glasgow Coma Scale — score from 3–15, measuring level of consciousness

RR	Respiratory Rate — breaths per minute
FiO2	Fraction of Inspired Oxygen — the percentage of oxygen a patient is receiving (room air = 21%)
SpO2	Oxygen Saturation — percentage of haemoglobin carrying oxygen, measured via pulse oximeter
SBAR	Situation, Background, Assessment, Recommendation — a structured format for clinical handover communication

Healthcare Systems & Infrastructure Terms

Term	Meaning
EHR	Electronic Health Record — digital patient record system; Mercer currently has none for triage, hence the Phase 1 proposal
LMIC	Low- and Middle-Income Country — a World Bank income classification used in global health literature
SIDS	Small Island Developing State — a UN classification for small island nations facing shared infrastructure constraints
Boarding time	The gap between a disposition decision (e.g. “admit”) and the patient physically leaving the ED, usually due to no ward bed being available

AI / Machine Learning & Risk Terms

Term	Meaning
AI	Artificial Intelligence
ML	Machine Learning
AUROC / AUC	Area Under the Receiver Operating Characteristic curve — a model performance metric from 0.5 to 1.0; higher is better
XGBoost	Extreme Gradient Boosting — a machine learning algorithm using an ensemble of decision trees, common for structured/tabular data
Rule-based classifier	A decision-making system using explicit if/then logic rather than a trained statistical model
PPV	Positive Predictive Value — the proportion of positive alerts that turn out to be true positives. Low PPV drives alert fatigue.
Automation bias	The tendency for a human user to over-trust or defer to an automated system's output, even when contradicted by their own judgment
Distribution shift	When a model's performance degrades because the data it encounters in deployment differs statistically from the data it was trained on

Human-in-the-loop	A system design principle requiring a human to review, confirm, or be able to override an AI-generated output before it takes effect
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Research & Documentation Terms

Term	Meaning
DOI	Digital Object Identifier — a permanent link format for academic papers
APA	American Psychological Association — the citation style used throughout this proposal (7th edition)
Scoping review	A literature review mapping the breadth of existing research, without necessarily assessing study quality in depth
Systematic review	A more rigorous literature review following a defined search and inclusion protocol